SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	K Suthesika	**Reg. No.:**	192511070
Section / Batch:	BE CSE	**Date:**	10/07/2026

Code of Conduct

I, K Suthesika (192511070) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K Suthesika

Instructions

- This test contains 20 Debugging & Optimization Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 -6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
		GRAND TOTAL:	**100 Marks**

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QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code - Linked List Node
10 marks

Q7
C Code - Time Complexity Loop
10 marks

Q1 C Code – Array Traversal

10 marks

```
#include <stdio.h>
int main() {
int arr[] = {5, 10, 15, 20};
int i, sum = 0;
for(i = 0; i < 4; i++)
sum += arr[i];
printf("%d", sum);
return 0;
}
```

Your Answer

50

Save Answer

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CO1_AT1_C CODE OUTPUT PREDICTION

Back

QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code - Linked List Node
10 marks

Q10 C Code - Pointer Increment

10 marks

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d",*p);
}
```

Your Answer

10

Save Answer